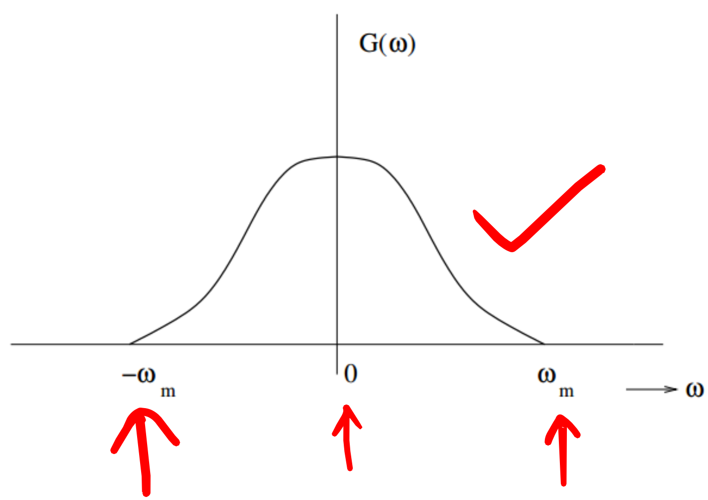
Chapter-5
Pulse Modulation and Demodulation

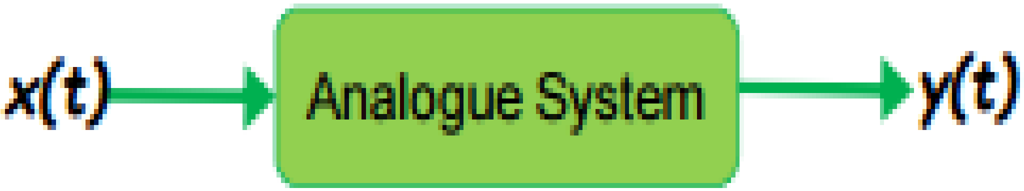
Complied By:
Kartik R. Patel
EXTC Department
KJSSE, SVU

Sampling of Low Pass Signal (Band limited)

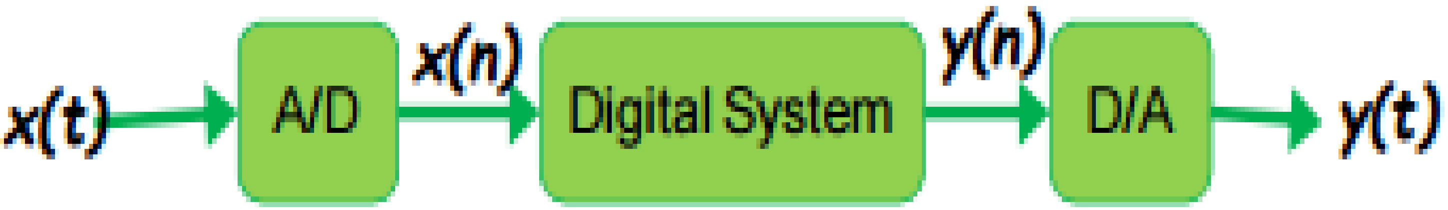


Continuous Versus Digital

Analogue electronic systems are continuous



Electronic System are increasingly digitalized



Signals are converted to numbers, processed, and converted back

Advantages of Digital over Analogue

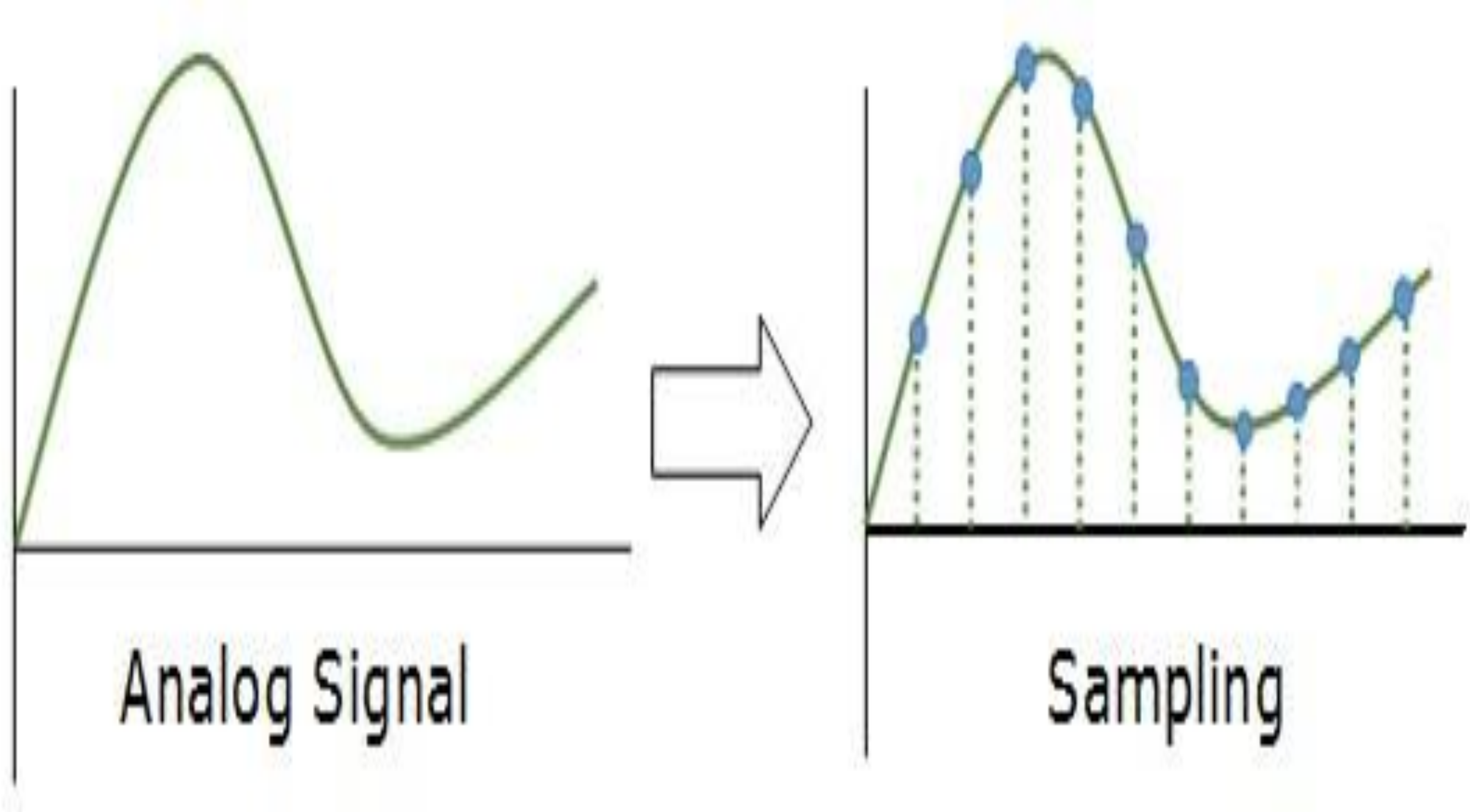
Advantages

- Flexibility (simply changing program)
- Accuracy
- Storage
- Ability to apply highly sophisticated algorithms.

Disadvantages

- It has certain limitations (very fast sample rate is needed when the bandwidth of signal is very large)
- It has a larger time delay compared to the analogue.

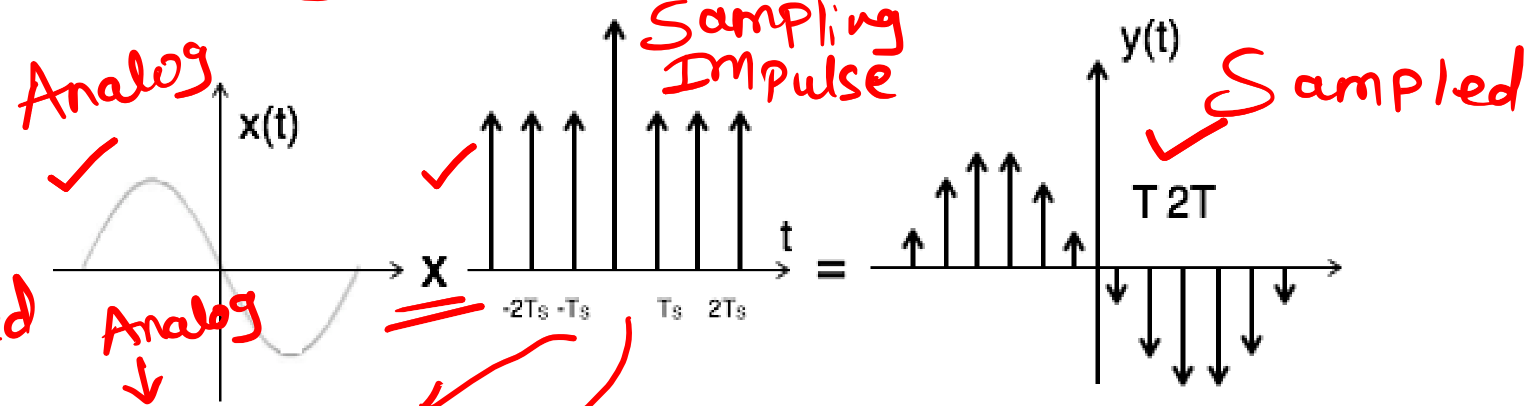
Analog to Digital conversion



Types of Sampling

- Impulse Sampling
- Natural Sampling
- Flat top Sampling

① Impulse Sampling



Sampled Analog

$$y(t) = x(t) \times \text{ impulse train}$$

$$= x(t) \times \Sigma_{n=-\infty}^{\infty} \delta(t - nT)$$

$n = 0$ $\delta(t)$

$n = 1$ $\delta(t - 1T)$, $n = -1$ $\delta(t + 1T)$

$$y(t) = y_\delta(t) = \Sigma_{n=-\infty}^{\infty} x(nt)\delta(t - nT) \ldots\ldots 1$$

To get the spectrum of sampled signal, consider Fourier transform of equation 1 on both sides

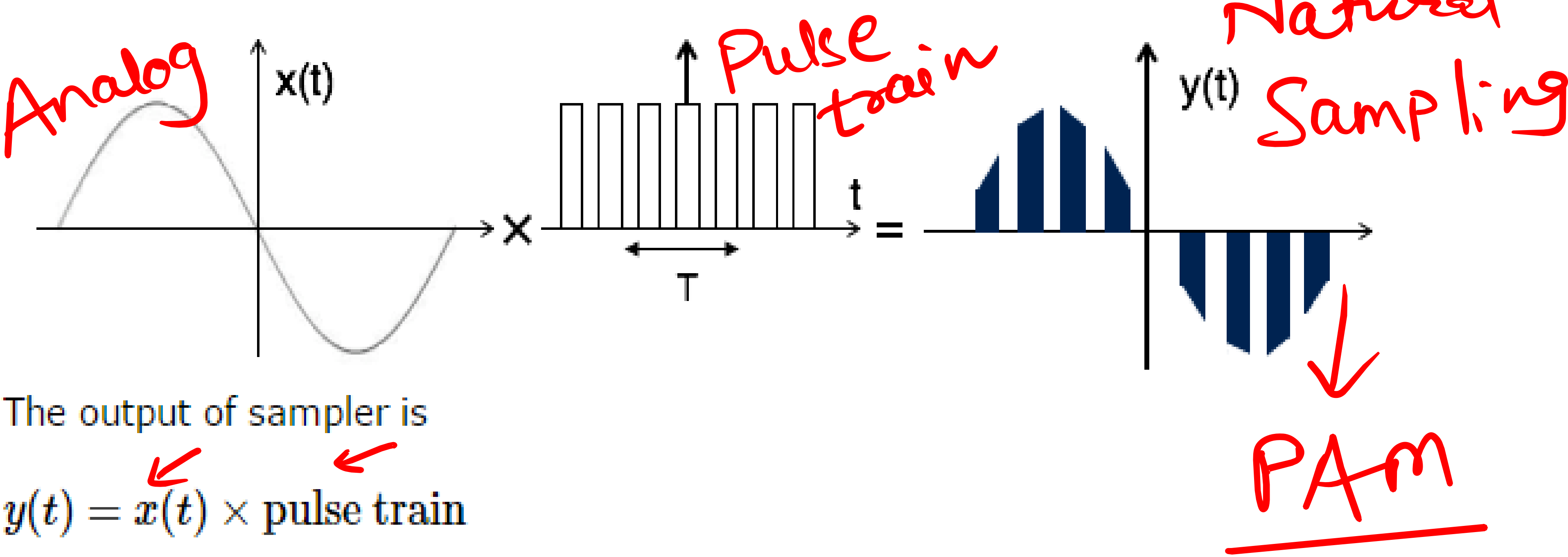
Frequency domain

$$Y(\omega) = \frac{1}{T}\Sigma_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

Derivation

2 Natural Sampling

Natural sampling is similar to impulse sampling, except the impulse train is replaced by pulse train of period T. i.e. you multiply input signal x(t) to pulse train $\Sigma_{n=-\infty}^{\infty} P(t-nT)$ as shown below



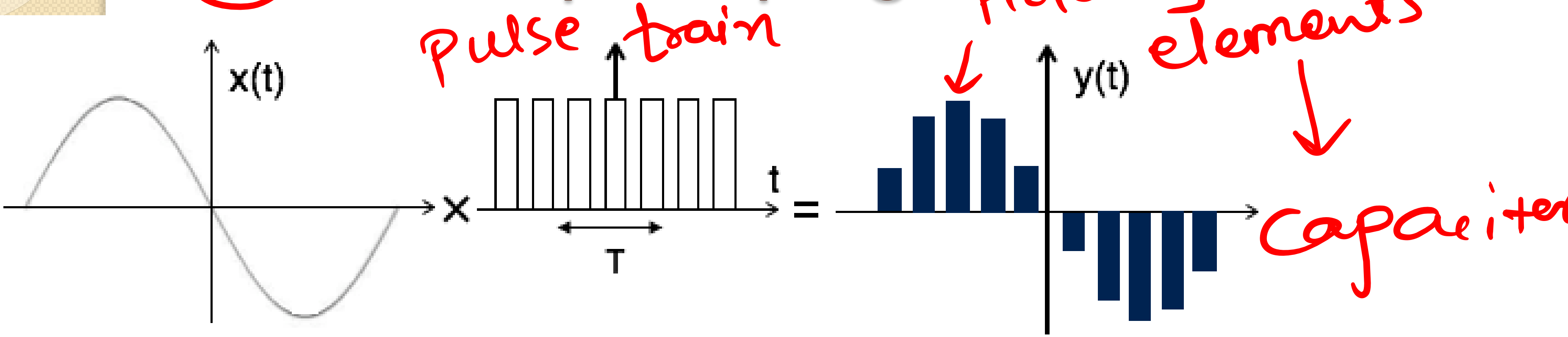
The output of sampler is

$$y(t) = x(t) \times \text{pulse train}$$

$$= x(t) \times p(t)$$

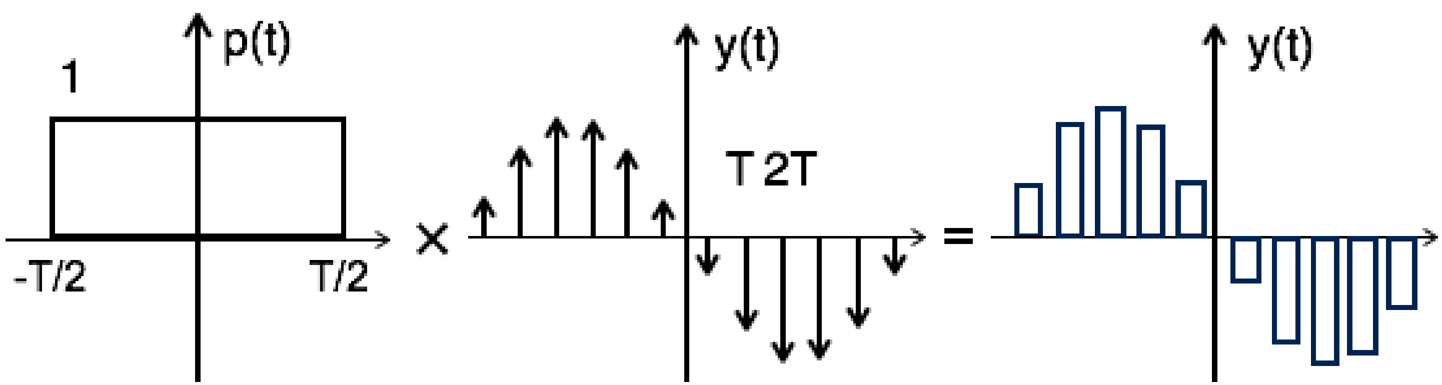
$$= x(t) \times \Sigma_{n=-\infty}^{\infty} P(t-nT) \ldots \ldots (1)$$

3 Flat top Sampling



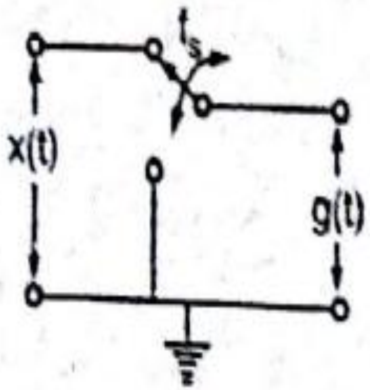
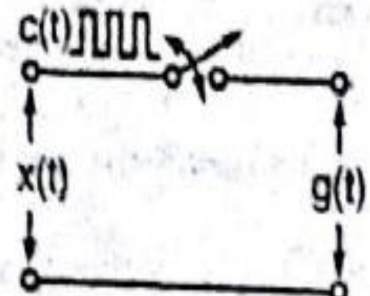
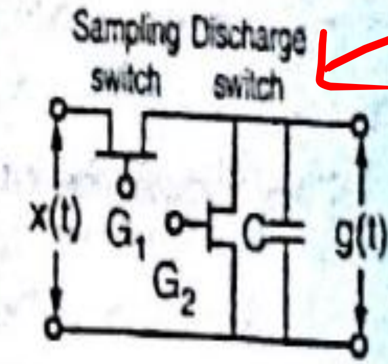
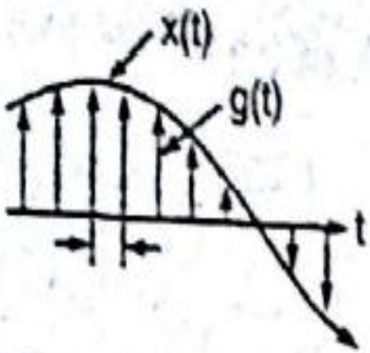
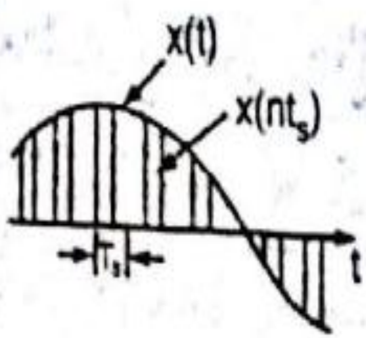
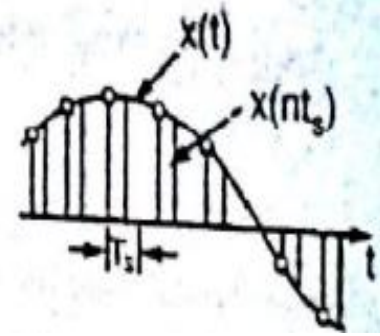
Theoretically, the sampled signal can be obtained by convolution of rectangular pulse p(t) with ideally sampled signal say $y_\delta(t)$ as shown in the diagram:

i.e. $y(t) = p(t) \times y_\delta(t) \ldots\ldots(1)$



Comparison

PAM

Parameter of comparison	Ideal or instantaneous sampling	Natural sampling	Flat top sampling
Sampling principle	It uses multiplication	It uses chopping principle	It uses sample and hold circuit
Generation circuit			PAM 
Waveforms involved			
Feasibility	This is not a		

Sampling Theorem

← Analog

"If a signal x(t) is sampled at regular intervals of time and at a rate higher than twice the highest signal frequency, then the samples contain all the information of the original signal."

$F_s > 2F_m$

The function x(t) may be reconstructed from these samples by the use of a low pass filter.

Sampling Theorem

- Sampling Theorem:

  "A bandlimited signal can be reconstructed exactly if it is sampled at a rate atleast twice the maximum frequency component in it."

Proof of Sampling Theorem

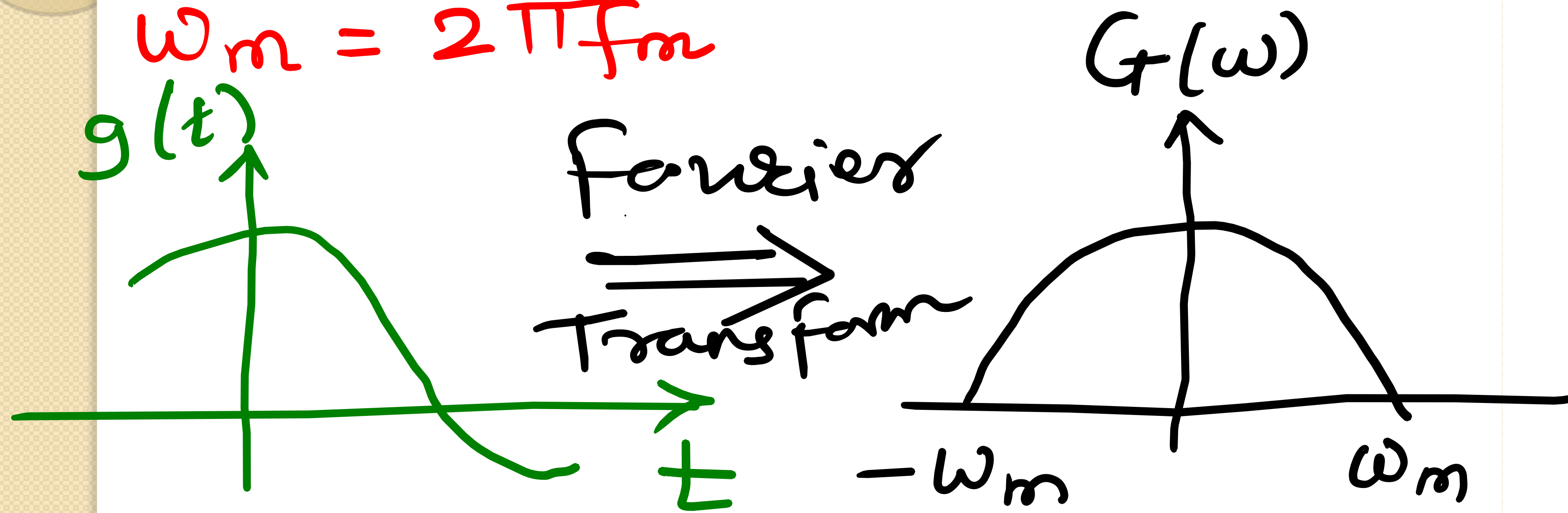
Analog signal $x(t)$

- The maximum frequency component of $g(t)$ is f_m. To recover the signal $g(t)$ exactly from its samples it has to be sampled at a rate $f_s \geq 2f_m$.
- The minimum required sampling rate $f_s = 2f_m$ is called *Nyquist rate.*

Proof of Sampling Theorem

Defining Analog signal g(t) and it's band limited signal

$\omega_m = 2\pi f_m$

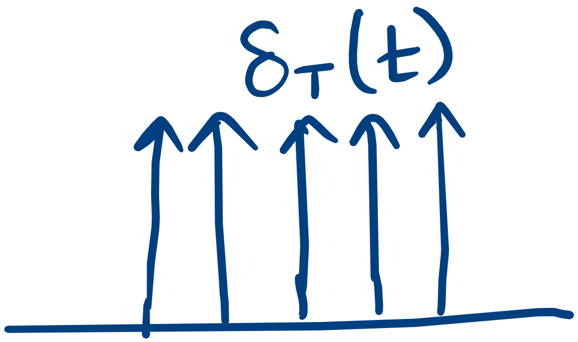


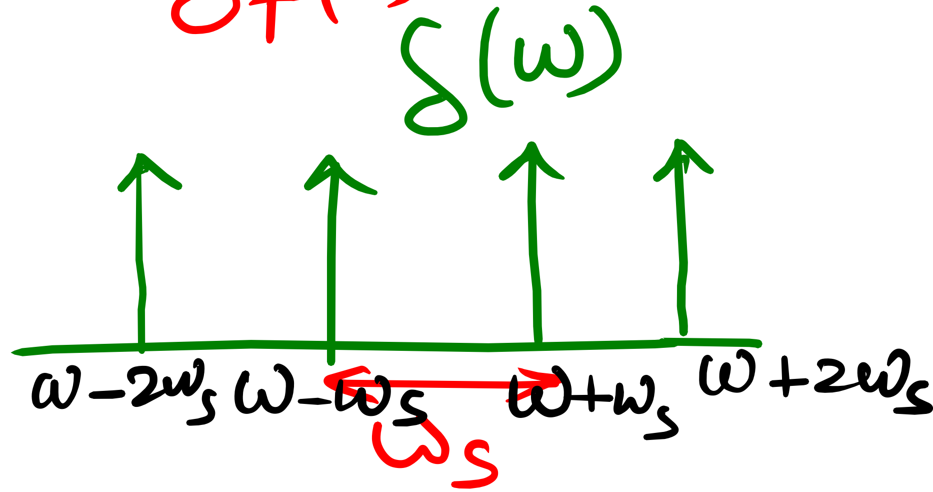
Proof of Sampling Theorem

Defining Sampling Signal $g_s(t)$ with period of $\frac{1}{T_s} \geq 2f_m$

$(f_s) \geq 2f_m$

→ Impulse Signal $\delta_T(t)$





Proof of Sampling Theorem

Sampled signal = Impulse x analog signal

$$g_s(t) = \delta(t) \times g(t) \quad -(1)$$

Taking Fourier transform of eq^n. (1)

$$F[g_s(t)] = F[\delta(t) \times g(t)] \quad -(2)$$

$$\delta(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Substitute $\delta(t)$ in eq^n. (2)

Proof of Sampling Theorem

Time domain $\rightarrow$ $F[g_s(t)] = F\left[g(t) \times \sum_{n=-\infty}^{\infty} \delta(t-nT)\right]$

Taking F.T

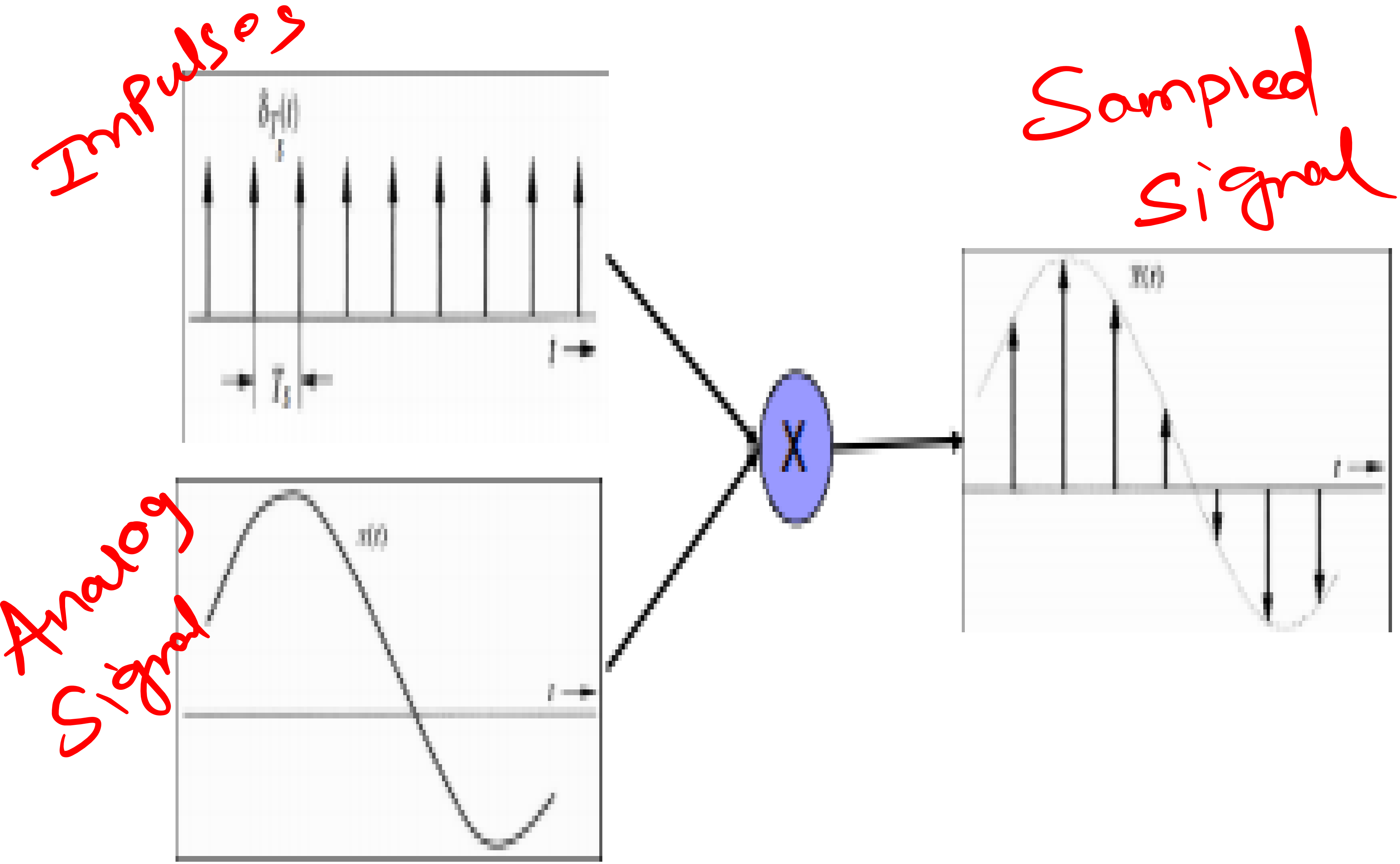
Frequency domain $\rightarrow$ $G_s(\omega) = \frac{1}{2\pi}\left[G(\omega) * \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)\right]$

$$\frac{\omega_0}{2\pi} = \frac{\cancel{2\pi} f_0}{\cancel{2\pi}} = f_0 = \frac{1}{T}$$

$$\boxed{G_s(\omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} G(\omega - n\omega_0)}$$

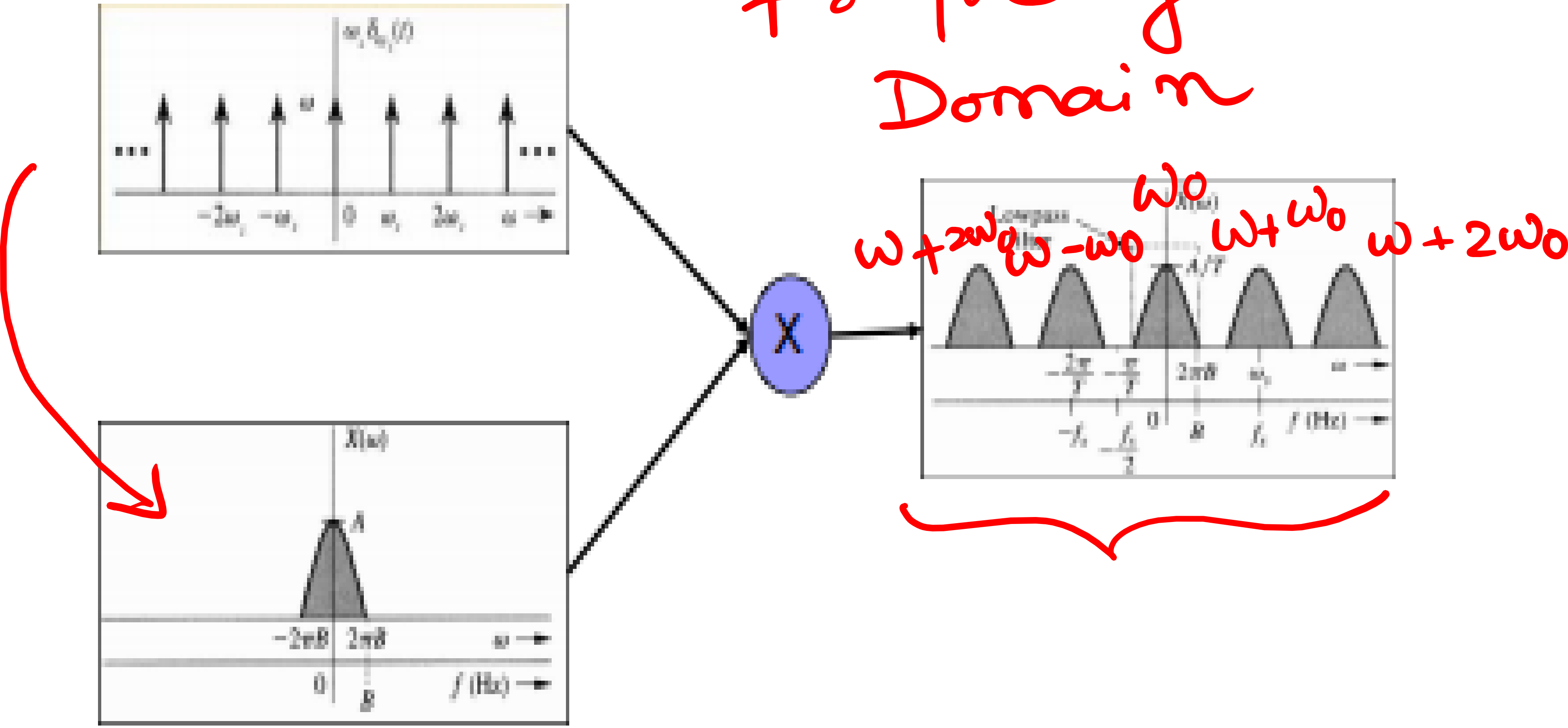
Sampled

Proof of Sampling Theorem



Proof of Sampling Theorem

Frequency
Domain



Sampling Theorem

Summary